

Master's Thesis — Selected Research Summary

Viscosity-Controlled Syringe Pump for Precise Filling of a Urological Implant with High-Viscosity Oils

Applicant	I-En Lee
Degree	M.Sc. Computational Engineering
Institution	Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)
Institute	Institute for Factory Automation and Production Systems (FAPS)
Supervisors	Prof. Dr. rer. nat. Jens Fürst; Alexander Preis, M.Sc.
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Full thesis	View full thesis on GitHub

Purpose of this document. This research summary presents the key aspects of my master's thesis. The selected material highlights my experience in precision fluid handling, mechanical design, physical modeling, embedded control, prototype development, and quantitative experimental validation. These capabilities provide a foundation for further research in biomedical devices, advanced manufacturing, systems integration, and model-based engineering.

Research snapshot

The thesis addressed precise syringe-based delivery of fluids spanning approximately **1–60,000 mPa·s** through a narrow cannula. I developed an integrated platform combining fluid-mechanical modeling, mechanical design, stepper-motor actuation, embedded control, a graphical user interface, viscosity-dependent compensation, and gravimetric validation.

Calibration points at 1, 5,000, 30,000, and 60,000 mPa·s were used to construct a monotonic interpolation map. At an intermediate **40,000 mPa·s** condition that was not used as a calibration point, all five validation trials remained within the defined $\pm 1\%$ volumetric tolerance, with a maximum absolute deviation of **0.93%**.

All images reproduced in this document are taken directly from the original master's thesis. Figure captions identify their corresponding source figure(s) in the submitted thesis. Explanatory text has been condensed and reorganized for PhD application purposes.

1. Research Problem and Engineering Objective

The application required repeatable filling of a small urological implant component using high-viscosity oil delivered through a narrow cannula. Under these conditions, commanded syringe-plunger displacement does not translate directly into delivered volume. Increasing viscosity and decreasing flow-path dimensions raise hydraulic resistance, while syringe compliance, plunger friction, structural deformation, and delayed material release can create additional deviations between actuator motion and outlet volume.

The engineering objective was therefore broader than designing a motorized syringe holder. The goal was to develop a process platform that could:

- relate target volume, fluid viscosity, syringe geometry, and cannula geometry to the filling process;
- provide mechanically stable and repeatable plunger actuation;
- convert process inputs into executable motion parameters;
- compensate for systematic viscosity-dependent deviations; and
- validate filling accuracy under controlled laboratory conditions.

The validation target used in the thesis was a nominal 1 mL delivery with a maximum permissible volumetric deviation of 1% under the defined experimental conditions.

2. Fluid-Mechanical Modeling and Numerical Evaluation

A simplified syringe–cannula model was used to describe how fluid properties and geometry influence the loading of the filling process. The model linked plunger displacement and velocity to volumetric flow and related the resulting cannula flow to pressure loss. For an idealized Newtonian flow in a cylindrical cannula, the first-order pressure estimate follows the Hagen–Poiseuille relation:

$$\Delta p_c = \frac{8\mu L_c Q}{\pi r_c^4},$$

where μ is dynamic viscosity, L_c is cannula length, Q is volumetric flow rate, and r_c is cannula inner radius.

The model was not treated as an exact transient pressure predictor. Instead, it provided a design-oriented description of the dominant dependencies that constrain actuator motion. **Python was used for numerical evaluation of the fluid-mechanical relationships and viscosity-dependent system behavior before implementation of the physical control strategy.**

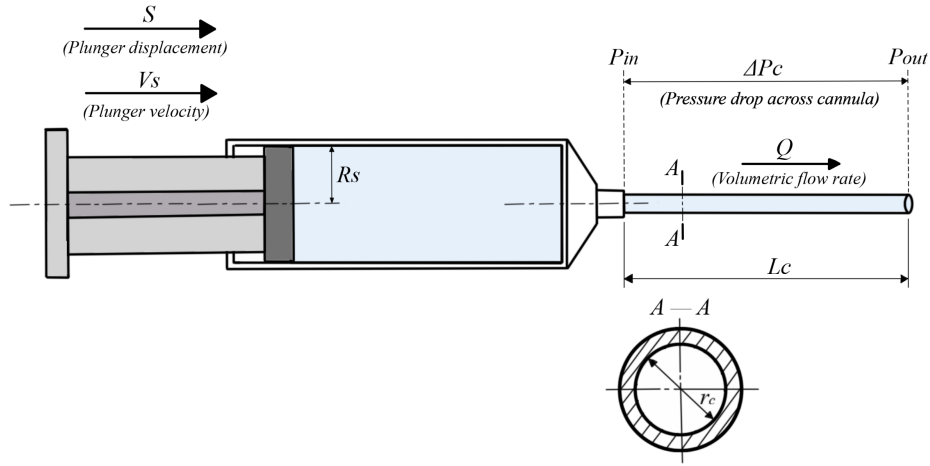


Figure 1 (original thesis Figure 5). Simplified syringe-cannula flow model used to describe the relationship between plunger motion, volumetric flow, cannula geometry, and pressure demand.

Two implications were particularly important for system design. First, viscosity changes pressure demand even when target volume and hardware geometry remain unchanged. Second, cannula radius has a strong nonlinear influence because it appears to the fourth power in the denominator.

For a fixed syringe and cannula geometry, increasing plunger velocity increases volumetric flow and therefore increases the estimated pressure demand. The actuator command consequently had to be treated as a **fluid-constrained motion-planning problem**, rather than only as a positioning task.

3. Mechanical Design and Manufacturability Considerations

The mechanical subsystem was developed around the disposable syringe, actuator, transmission, and guidance requirements. Particular attention was given to axial alignment of the actuator force with the syringe axis, repeatable syringe positioning, replaceable printed parts, practical syringe exchange, and interfaces that could accommodate the dimensional limitations of FDM-printed components.

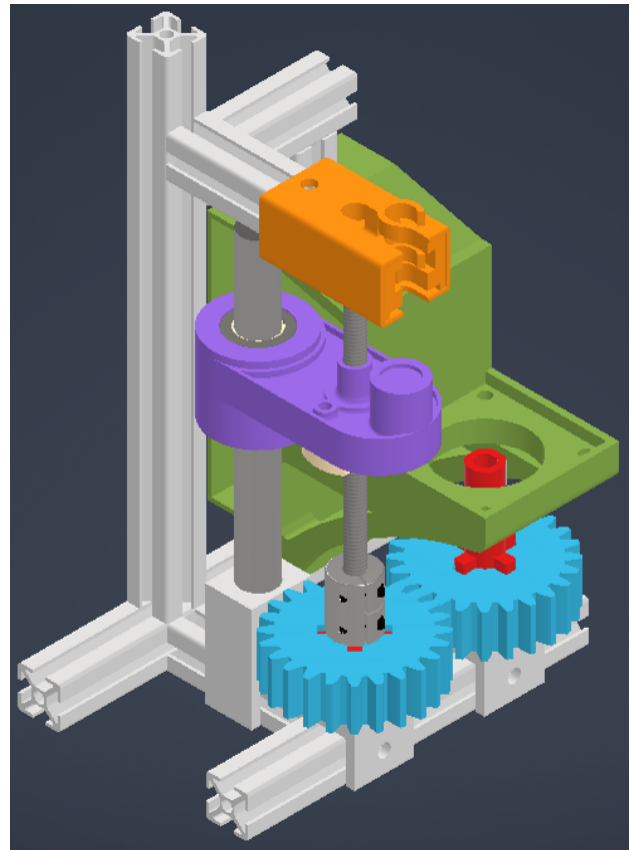
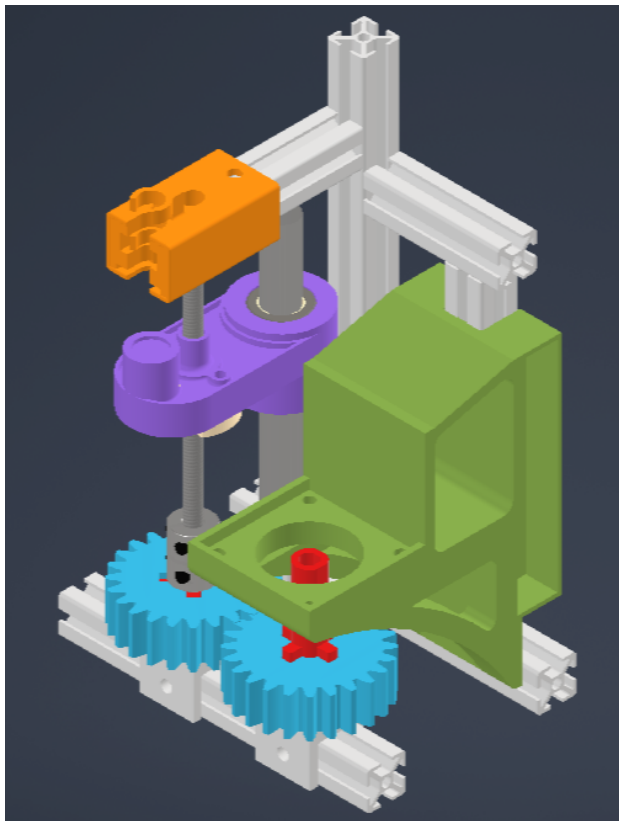


Figure 2 (original thesis Figure 17). Mechanical subsystem shown from two CAD perspectives, illustrating the structural frame, syringe fixation, plunger-contact interface, guidance elements, actuator mounting, and transmission system.

The syringe fixation used a quick-release concept and semi-enclosing support to improve repeatable positioning while avoiding unnecessary over-constraint. The plunger-contact interface incorporated clearance to accommodate small alignment and manufacturing deviations.

The design process considered how components would actually be produced and assembled. This included printed-part geometry, dimensional tolerances, clearance for assembly, modular interfaces, replaceable components, and practical assembly sequences. These considerations supported both manufacturability and repeatable laboratory assembly.

4. Prototype Realization, Actuation, and Embedded Control

The mechanical concept was developed into a test-ready electromechanical prototype through CAD refinement, component fabrication, assembly, actuator integration, and verification that the built hardware matched the intended operating concept.

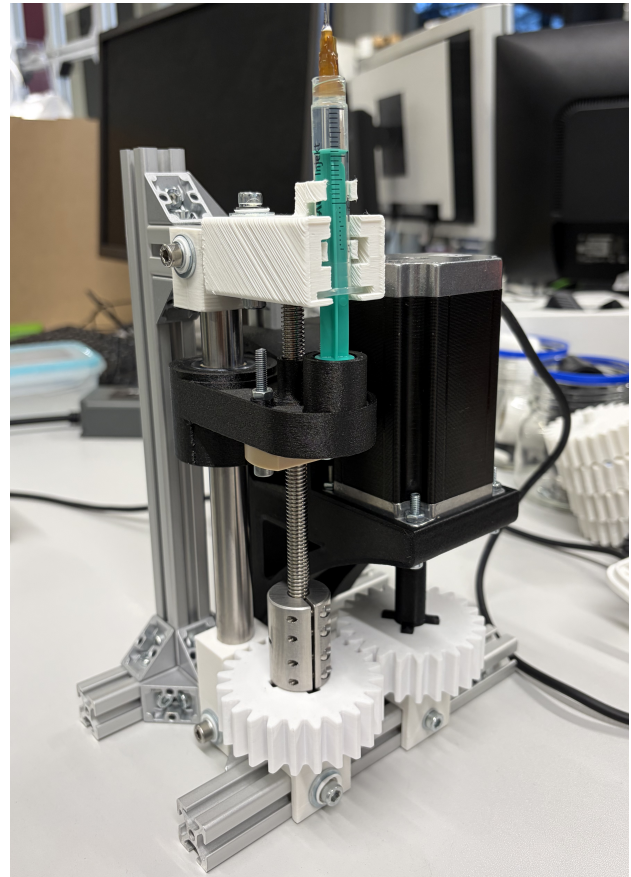
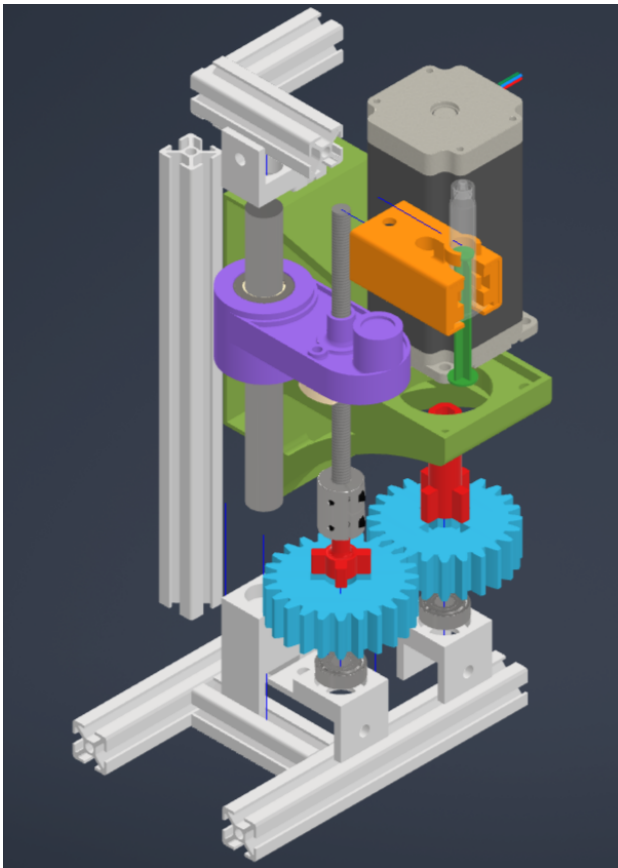


Figure 3 (from original thesis Figure 39). Left: CAD-based installation reference of the developed system. Right: assembled test-ready syringe-pump prototype used during experimental validation.

The final platform integrated a syringe fixture, leadscrew-based linear actuation, printed transmission components, supporting frame, and control electronics into a usable laboratory system.

The actuation system used a **NEMA 23 stepper motor** and a **custom TMC5160-TA-based motor-driver board**. An **Arduino GIGA R1 WiFi** served as the main embedded controller. Embedded **C/C++** software handled parameter processing, motion execution, driver communication, process monitoring, and graphical user interaction. Motor-side status monitoring also incorporated **INA228-based electrical measurements**.

This created a complete model-to-hardware workflow: **Python-based numerical evaluation** → **actuator and mechanical design** → **embedded C/C++ implementation** → **physical experimental testing**.

5. Compensation Strategy and Viscosity-Dependent Interpolation

The final control approach was implemented as model-based open-loop extrusion control. Target volume and viscosity were converted into a distance-based actuator command, constrained by the modeled operating range and corrected using experimentally determined compensation parameters.

Direct pressure and outlet-flow feedback were not available in the prototype. Systematic deviations were therefore addressed before motion execution through calibration-based compensation and viscosity-dependent parameter estimation.

The effective motion scale was selected as the main compensation parameter. Calibration points were established at

1, 5,000, 30,000, and 60,000 mPa·s. Because the viscosity range spans several orders of magnitude, the compensation map was constructed in the logarithmic viscosity domain using monotonic interpolation.

The intermediate 40,000 mPa·s condition was reserved as a validation condition rather than used as another manually tuned calibration point.

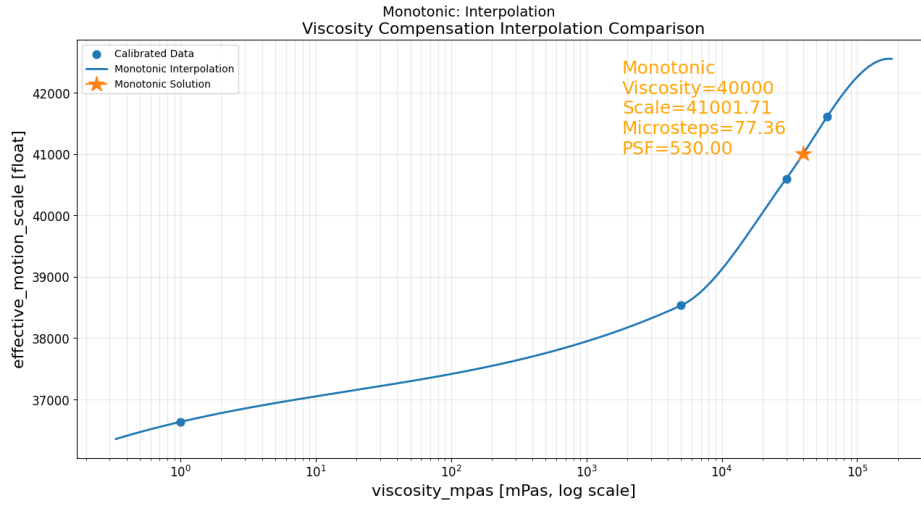


Figure 4 (original thesis Figure 44). Monotonic interpolation of the viscosity-dependent effective motion scale. The interpolation predicted an effective motion scale of 41,001.71 for the 40,000 mPa·s validation condition.

The interpolation complemented the physical model by compensating for systematic differences between ideal actuator displacement and actual delivered output under different viscosity conditions. Experimentally calibrated anchor points were therefore used to estimate a compensation parameter for intermediate viscosities within the tested range.

6. Experimental Validation and Key Result

The oil-based validation workflow used a fixed syringe–cannula configuration and controlled laboratory procedures to reduce avoidable trial-to-trial variation. The experiments were conducted at approximately 23°C using the same B. Braun Injekt Luer Lock Solo 2 mL syringe type and fixed cannula configuration. Syringes were used once, and the KERN PCD balance was tared before measurement.

For the high-viscosity oils, delivered volume was determined gravimetrically from the difference between the **syringe mass before and after extrusion**. The measured mass difference was converted to delivered volume using the **supplier-specified specific gravity** and compared with the nominal 1 mL target.

After extrusion, the plunger-contact interface was retracted and a standardized **3-min post-release waiting interval** was applied before evaluation.

The 40,000 mPa·s condition had not been included among the four calibration anchor points. The interpolated compensation value was applied without additional manual retuning, and five repeated validation trials were performed.

Condensed from original thesis Table 11. Compensated extrusion performance at 40,000 mPa·s.

Trial	Syringe mass before [g]	Syringe mass after [g]	Volumetric deviation [%]
1	4.334	3.358	0.62
2	4.294	3.331	-0.72
3	4.328	3.366	0.82
4	4.268	3.307	-0.93
5	4.297	3.322	0.52
Mean absolute deviation			0.72
Maximum absolute deviation			0.93

All five trials remained within the defined $\pm 1\%$ volumetric tolerance. The maximum absolute deviation was **0.93%**, and the mean absolute deviation was **0.72%**.

Engineering significance

The result demonstrates a complete research workflow: **physical modeling** → **numerical evaluation** → **device design** → **embedded implementation** → **experimental calibration** → **prediction at an intermediate condition not used for calibration** → **quantitative validation**.

The experiments also showed that actuator displacement alone did not fully describe post-extrusion behavior. For high-viscosity oils, residual pressure and delayed outflow remained relevant after commanded motion ended. A passive-release step was therefore introduced by retracting the plunger-contact interface after extrusion.

This observation reinforced the need to treat the system as a coupled **actuator–structure–fluid system** rather than as a purely kinematic mechanism.

7. Transferable Research Capabilities

This thesis developed research capabilities that are directly relevant to biomedical device and systems research and advanced manufacturing, particularly where fluid behavior, mechanical design, actuation, fabrication, and experimental validation must be considered together:

- **Fluid-mechanical reasoning:** relating fluid properties and flow-path geometry to pressure demand, actuator loading, and device response.
- **Python-based numerical modeling:** evaluating fluid-dependent system behavior and candidate operating conditions before physical implementation.
- **Model-to-hardware translation:** converting physical and numerical analysis into actuator parameters, embedded control, and a working prototype.
- **Electronic and embedded-system integration:** integrating stepper-motor actuation, a custom TMC5160-TA-based driver, Arduino GIGA hardware, embedded C/C++, and process monitoring.
- **Mechanical design and prototyping:** CAD development, tolerance-aware interfaces, modular assembly, additive manufacturing, and iterative physical refinement.
- **Quantitative experimental validation:** defining measurable performance criteria, conducting repeated controlled trials, identifying systematic deviations, calibrating the system, and validating an intermediate condition not used for calibration.

These capabilities are directly transferable to biomedical device and systems research in which device geometry, mechanical behavior, fluid transport, fabrication, and experimentally measured performance must be considered together. This research experience provides a foundation for further development of biomedical devices and integrated engineering systems, while allowing me to continue expanding my expertise in advanced manufacturing, systems integration, and experimental validation.